YAN DVORETSKIY

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EDUCATION

Dartmouth College, Hanover, NH — **Aug 2022 – Jun 2026**

Bachelor of Arts (B.A.) in Computer Science modified with Engineering Science — **GPA: 3.5/40**

Bachelor of Engineering (B.E.) in Computer Engineering

Relevant Coursework: *Embedded Systems, Digital Electronics, Microprocessors, Computer Architecture, Operating Systems*

EXPERIENCE

Wilcox Industries, Newington, NH — **Jun 2025 – Aug 2025**

Firmware / Electrical Engineering Intern

- Designed, prototyped, and debugged a modular embedded software platform on STM32 for scalable sensor testing across current and future product lines reducing test setup from hours to minutes
- Created a custom PCB in Altium and led hardware bring-up, validation, and firmware integration, which enabled repeatable testing across peripherals and eliminated an entire class of manual regression errors
- Developed embedded firmware in C++ with modular, layered abstractions (drivers, interfaces, CLI), enabling extensibility and reuse across 3+ hardware variants
- Built Python + Qt tooling for real-time visualization and analysis of sensor response, enabling rapid identification of calibration errors and reducing debugging effort during validation.
- Contributed across the full engineering lifecycle (requirements, schematic/PCB reviews, implementation, validation, and documentation) within Git-based workflows on large legacy codebases

Thayer School of Engineering at Dartmouth, Hanover, NH — **Jun 2024 – Aug 2024**

Teaching Assistant, Digital Electronics

- Supported 40+ students in logic design, state machines, FPGA programming, and debugging complex hardware/software interactions
- Led weekly help sessions and 1:1 instruction, translating abstract technical concepts into clear, actionable explanations

PROJECTS

9 Axis Attitude Estimation System (C++)

- Developed bare-metal C++ firmware on ARM Cortex-M STM32 using SPI/I²C interfaces and interrupt-driven sensor acquisition.
- Implemented a real-time Extended Kalman Filter and supporting math routines from scratch fusing accelerometer, gyroscope, and magnetometer data for 3D orientation estimation using quaternion representation
- Built a Python simulation framework generating synthetic sensor data from known orientation sequences to validate filter accuracy without physical data collection

Ski Performance & Injury Prevention Wearable (C++)

- Built the full hardware-to-software stack for a sponsor-funded ESP32 wearable, integrating an IMU, force sensor, and haptic/LED feedback into real-time coaching for beginner-to-intermediate skiers
- Implemented FreeRTOS firmware in C++ using Madgwick sensor fusion for orientation and dual-core parallelism to stream data to a companion app; achieved 90% turn-detection accuracy
- Debugged and patched open-source sensor libraries against datasheets, upstreaming fixes accepted by maintainers

Yalnix Operating System (C)

- Built a Unix-like operating system featuring Round-Robin scheduling, virtual memory (page tables/TLB), context switching, and interrupt handling. Designed system calls and terminal I/O drivers to support concurrent process execution, memory allocation, file operations, and multi-terminal access.

LEADERSHIP & EXTRACURRICULAR

Dartmouth Swimming and Diving, Hanover, NH — **Aug 2022 – Jun 2026**

Varsity Swim Team Captain

- Balanced 20+ hours/week of Division I athletics with a rigorous dual degree engineering curriculum
- Elected Team Captain for two consecutive years; served as primary liaison between 50 athlete roster and coaching staff, coordinating schedules and team-wide initiatives
- Won a national championship in the 200-yard butterfly and set numerous program records, demonstrating sustained discipline, accountability, and performance under pressure

TECHNICAL SKILLS

Programming: C++, C, Rust, Python, VHDL, MATLAB, SQL, Git, Linux
Embedded & HW: STM32, ESP32, FreeRTOS, SPI, I2C, UART, Oscilloscopes, Logic Analyzers, Altium Designer, Vivado
Languages: English (Fluent), Russian (Fluent)